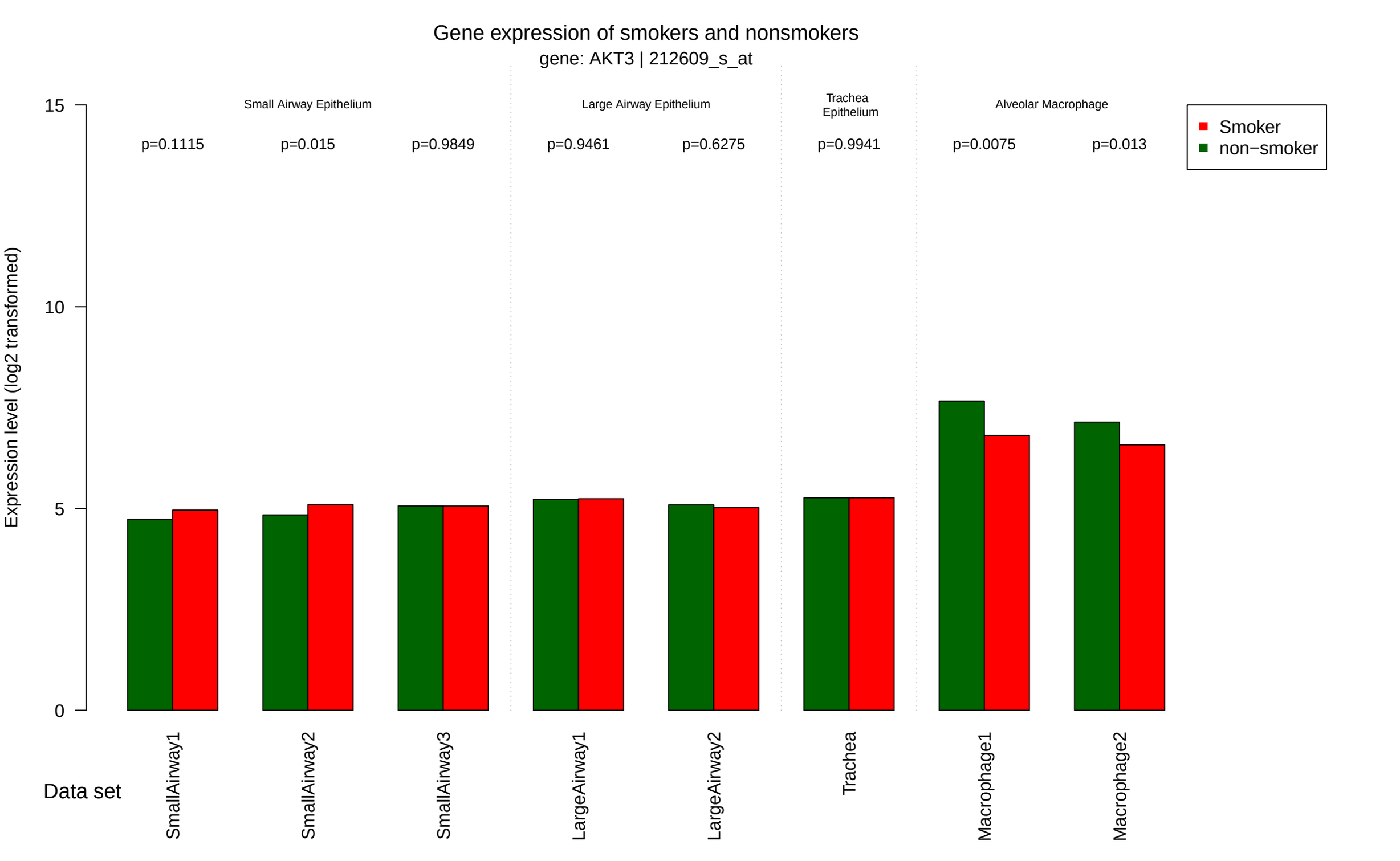
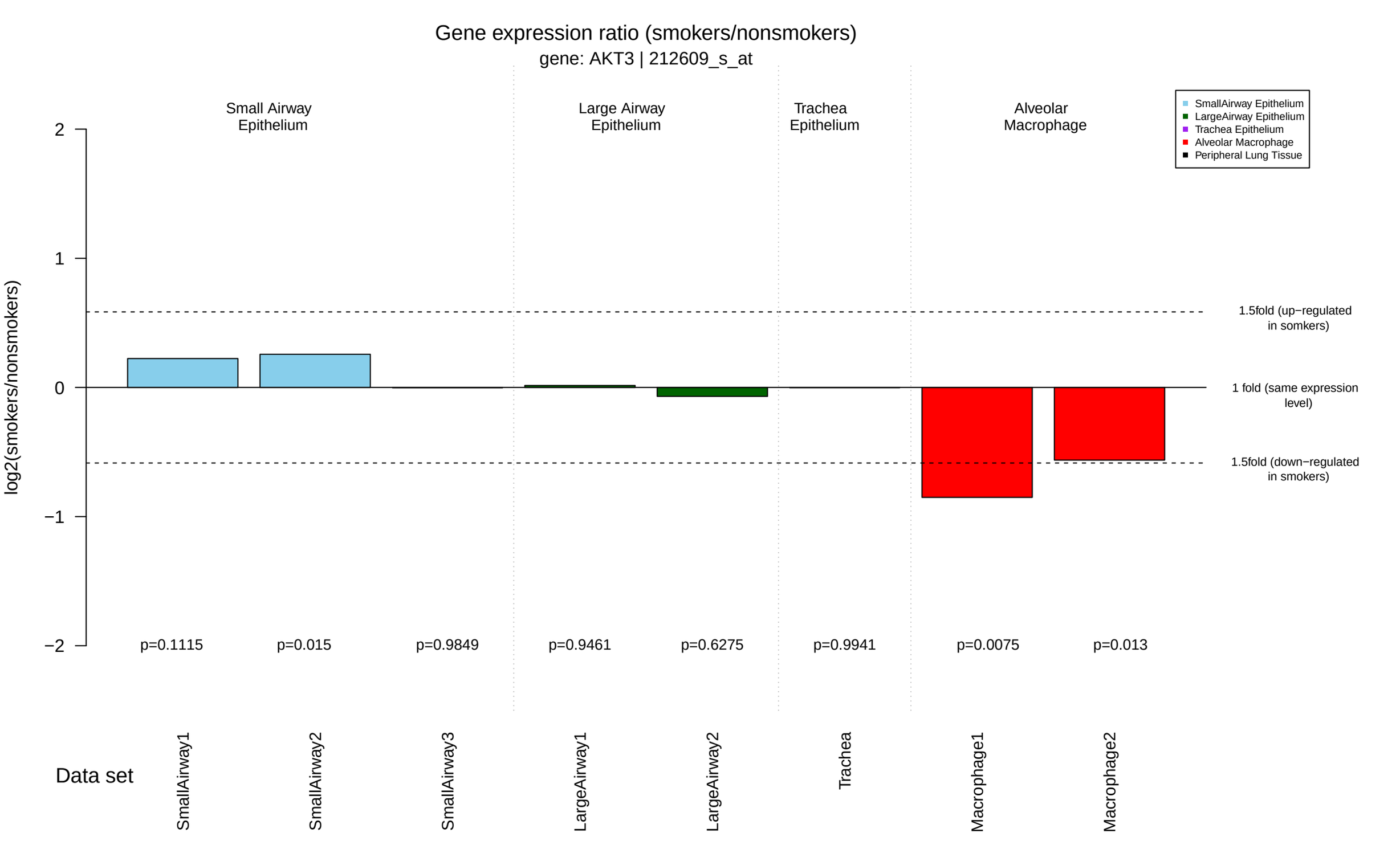
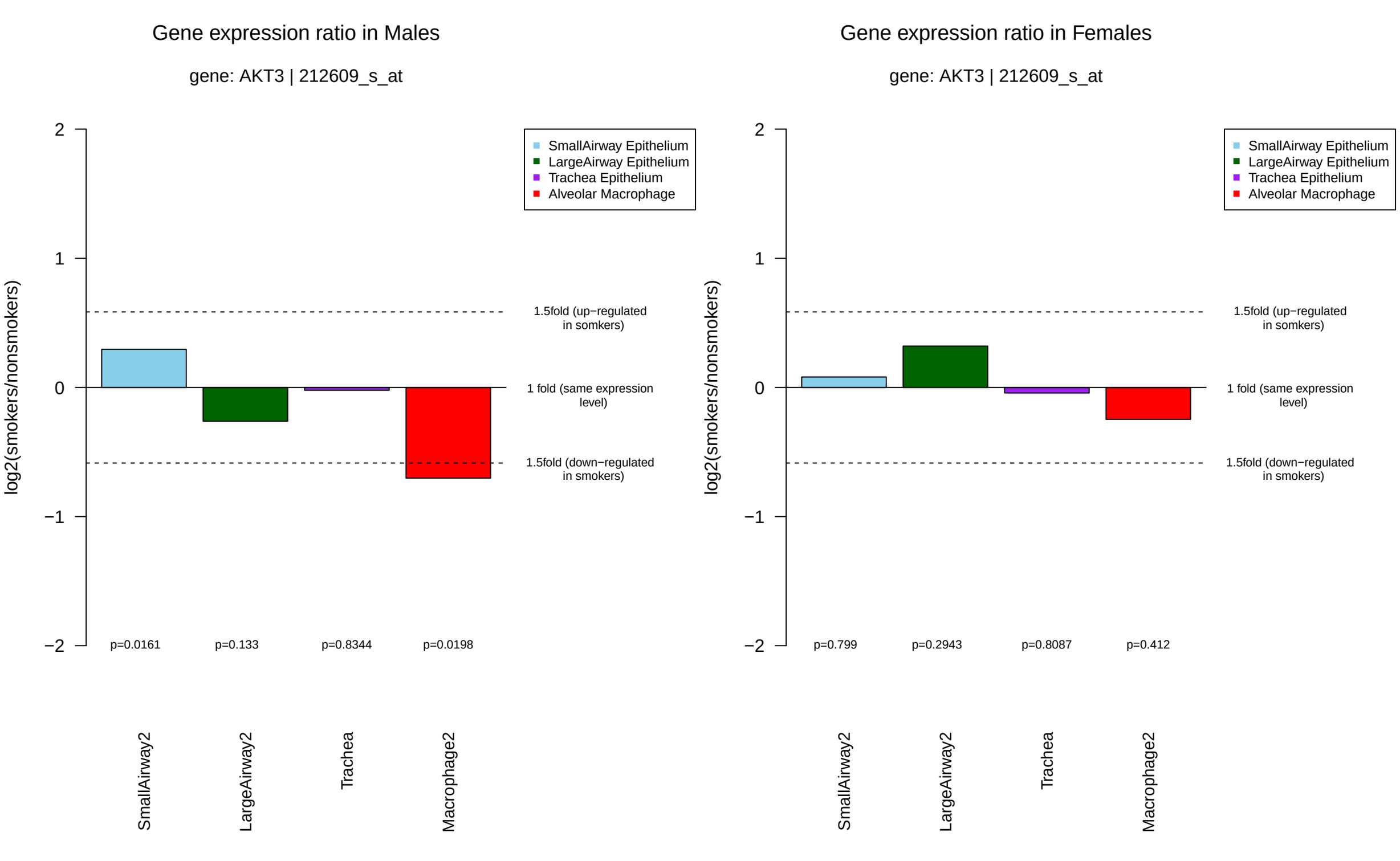
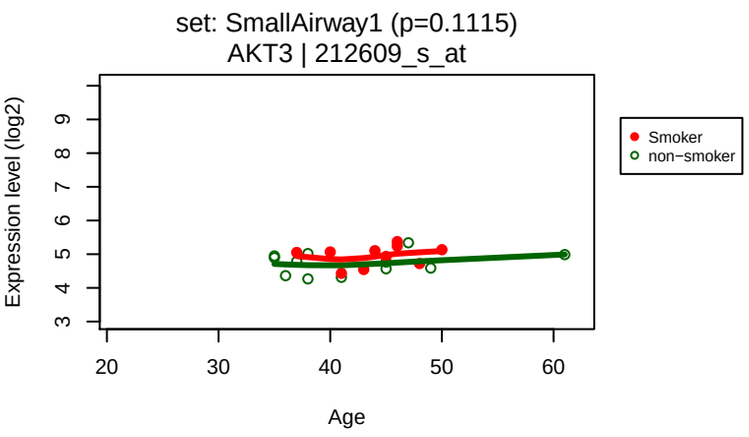
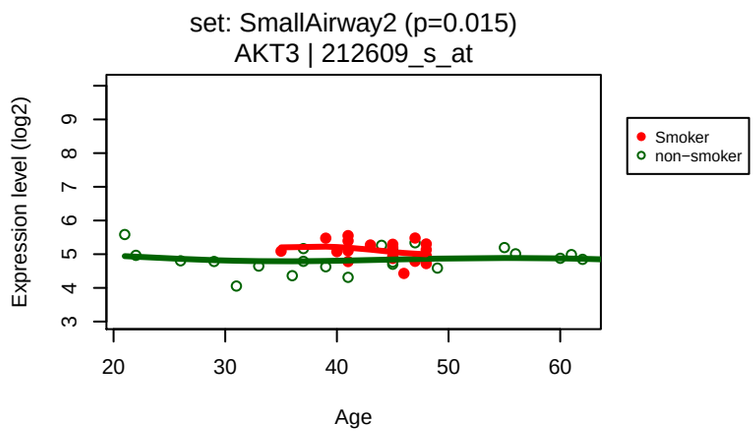
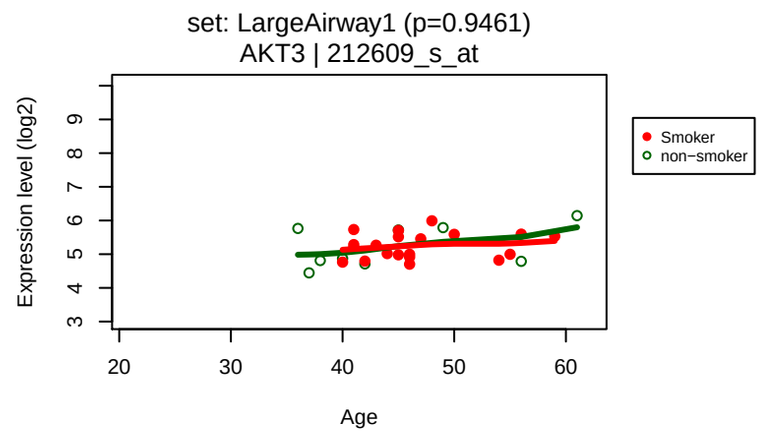

Gene expression of smokers and nonsmokers
gene: AKT3 | 212609_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1115
p=0.015
p=0.9849
p=0.9461
p=0.6275
p=0.9941
p=0.0075
p=0.013
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

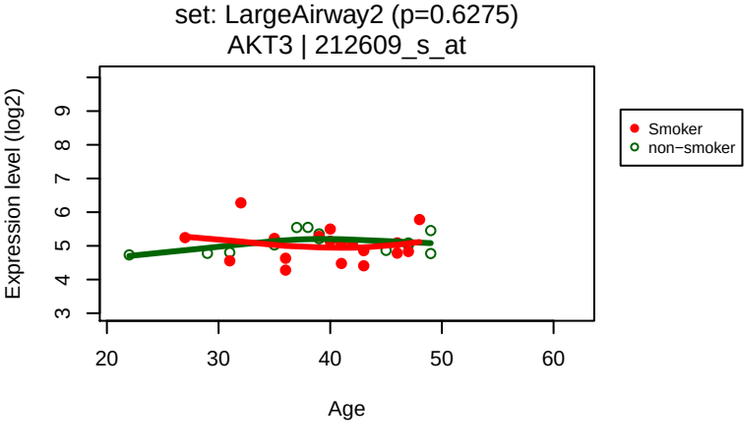

Gene expression ratio (smokers/nonsmokers)
gene: AKT3 | 212609_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1115
p=0.015
p=0.9849
p=0.9461
p=0.6275
p=0.9941
p=0.0075
p=0.013
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

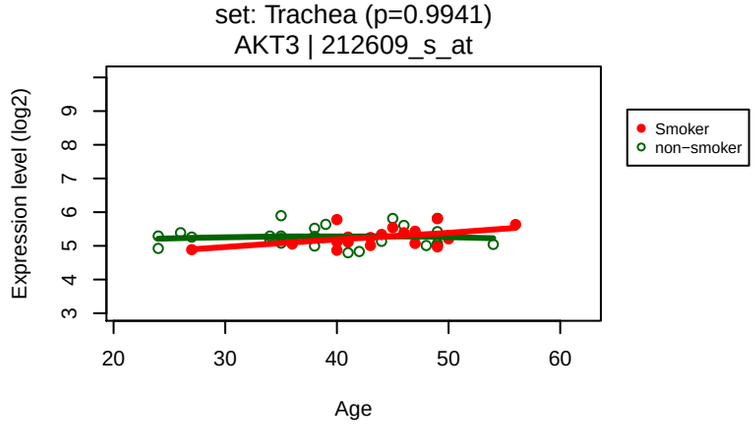

Gene expression ratio in Males
gene: AKT3 | 212609_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0161
p=0.133
p=0.8344
p=0.0198
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKT3 | 212609_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.799
p=0.2943
p=0.8087
p=0.412
SmallAirway2
LargeAirway2
Trachea
Macrophage2

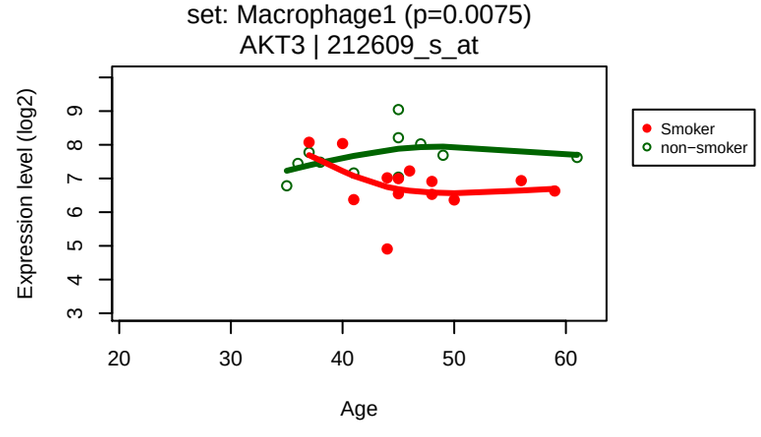

set: SmallAirway1 (p=0.1115)
AKT3 | 212609_s_at
Expression level (log2)
Age
Smoker
non−smoker

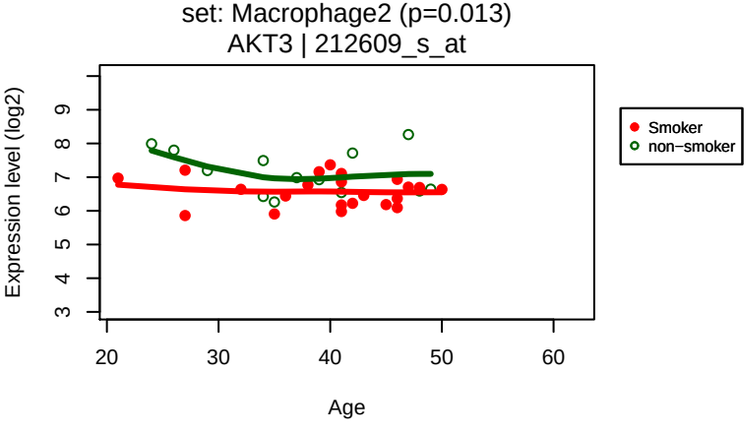

set: SmallAirway2 (p=0.015)
AKT3 | 212609_s_at
Expression level (log2)
Age
Smoker
non−smoker

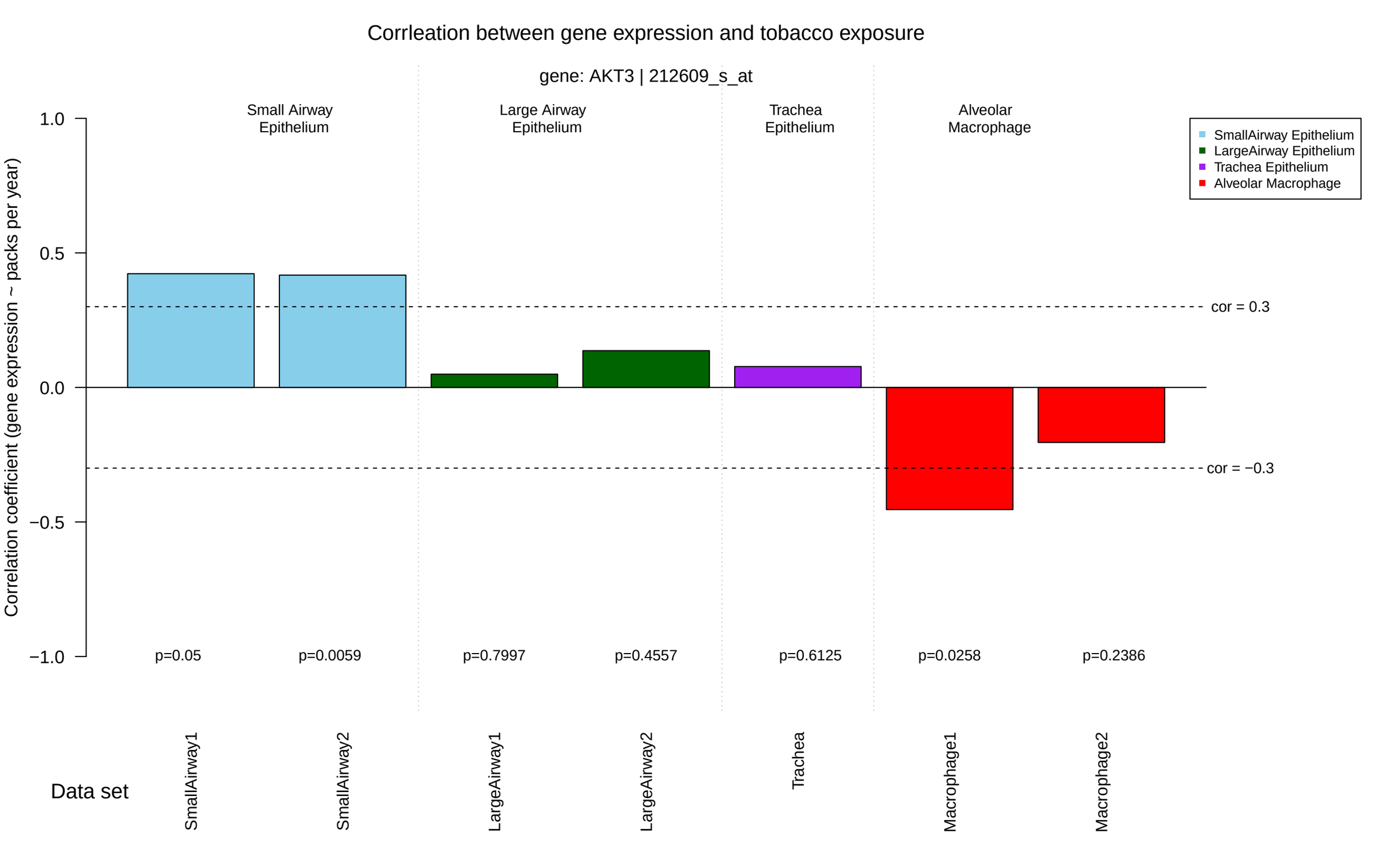

set: LargeAirway1 (p=0.9461)
AKT3 | 212609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6275)
AKT3 | 212609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9941)
AKT3 | 212609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0075)
AKT3 | 212609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.013)
AKT3 | 212609_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKT3 | 212609_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.05
p=0.0059
p=0.7997
p=0.4557
p=0.6125
p=0.0258
p=0.2386
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2